MATH 181A - Homework 2

due Saturday, January 17 at 11:59PM

Problem 1.

You need to do the MLE in 5.2.12 as well to make a comparison.

1.1 5.2.12 A random sample of size n is taken from the pdf

$$f_Y(y;\theta) = 2y/\theta^2, \quad 0 \le y \le \theta$$

Find an expression for $\hat{\theta}$, the maximum likelihood estimator for θ.

Solution:

$$L(\theta) = \prod_{i=1}^{n} f_Y(y_i;\theta) = \prod_{i=1}^{n} \frac{2y_i}{\theta^2}$$

$$\frac{d}{d\theta}L(\theta) = \frac{d}{d\theta}\left(\frac{2^n}{\theta^{2n}}\prod_{i=1}^{n} y_i\right) = -\frac{2^{n+1}n}{\theta^{2n+1}}\prod_{i=1}^{n} y_i < 0 \qquad \forall\theta \ge 0$$

Since the likelihood function is strictly decreasing and we know that $\theta \ge y_i$ for all i, the maximum likelihood estimator is the smallest number $\ge y_i$ for all i. Thus, $\hat{\theta} = \max_{i=1}^{n} y_i$.

1.2 5.2.17 Let $y_1, y_2, \ldots, y_n$ be a random sample of size n from the pdf $f_Y(y;\theta) = 2y/\theta^2$, $0 \le y \le \theta$. Find a formula for the method of moments estimate for θ. Compare the values of the method of moments estimate and the maximum likelihood estimate if a random sample of size 5 consists of the numbers 17, 92, 46, 39, and 56 (recall Question 5.2.12).

Solution:

$$\mathbb{E}[Y] = \int_0^\theta y\frac{2y}{\theta^2}dy = \frac{2}{\theta^2}\int_0^\theta y^2 dy = \frac{2}{\theta^2}\left[\frac{y^3}{3}\right]_0^\theta = \frac{2\theta^3}{3\theta^2} = \frac{2\theta}{3}$$

$$\frac{2\theta}{3} = \bar{y} \implies \theta = \frac{3}{2}\bar{y}$$

Thus, $\theta_e = \frac{3}{2}\bar{y}$. Comparing this to the maximum likelihood estimator, we get $\hat{\theta} = \max_{i=1}^{n} y_i = 92$ and $\theta_e = \frac{3}{2}\cdot\frac{17+92+46+39+56}{5} = 75$ when the sample is (17, 92, 46, 39, 56). We can see here that the two estimates are different and generally won't give the same results.

Problem 2. 5.2.18

Use the method of moments to estimate θ in the pdf $f_Y(y;\theta) = (\theta^2+\theta)y^{\theta-1}(1-y)$, $0 \le y \le 1$. Assume that a random sample of size n has been collected.

Solution:

$$\begin{aligned}
\mathbb{E}[Y] &= \int_0^1 y(\theta^2+\theta)y^{\theta-1}(1-y)dy = \int_0^1 (\theta^2+\theta)y^{\theta}(1-y)dy \\
&= \int_0^1 (\theta^2+\theta)y^{\theta} - (\theta^2+\theta)y^{\theta+1}dy \\
&= (\theta^2+\theta)\int_0^1 y^{\theta}dy - (\theta^2+\theta)\int_0^1 y^{\theta+1}dy \\
&= (\theta^2+\theta)\left[\frac{y^{\theta+1}}{\theta+1}\right]_0^1 - (\theta^2+\theta)\left[\frac{y^{\theta+2}}{\theta+2}\right]_0^1 \\
&= (\theta^2+\theta)\left(\frac{1}{\theta+1}\right) - (\theta^2+\theta)\left(\frac{1}{\theta+2}\right) \\
&= \theta - \frac{\theta^2+\theta}{\theta+2} = \frac{\theta}{\theta+2} = \bar{y} \\
&\implies \theta_e = \frac{2\bar{y}}{1-\bar{y}}
\end{aligned}$$

Where $\bar{y} = \sum_{i=1}^n y_i/n$.

Problem 3. 5.2.21

Suppose that $Y_1 = 8.3, Y_2 = 4.9, Y_3 = 2.6, and Y_4 = 6.5$ is a random sample of size 4 from the two-parameter uniform pdf,

$$f_Y(y; \theta_1, \theta_2) = 1/(2\theta_2), \quad \theta_1 - \theta_2 \leq y \leq \theta_1 + \theta_2$$

Use the method of moments to calculate θ_{1e} and θ_{2e}.

Solution:

$$\begin{aligned}
\mathbb{E}[Y] &= \int_{\theta_1-\theta_2}^{\theta_1+\theta_2} y \frac{1}{2\theta_2} dy = \frac{1}{2\theta_2} \int_{\theta_1-\theta_2}^{\theta_1+\theta_2} y dy = \frac{1}{2\theta_2} \left[\frac{y^2}{2}\right]_{\theta_1-\theta_2}^{\theta_1+\theta_2} \\
&= \frac{1}{2\theta_2} \left(\frac{\theta_1^2 + 2\theta_1\theta_2 + \theta_2^2}{2} - \frac{\theta_1^2 - 2\theta_1\theta_2 + \theta_2^2}{2} \right) \\
&= \frac{1}{2\theta_2} \left(\frac{4\theta_1\theta_2}{2} \right) = \theta_1 = \bar{y} \\
&\implies \theta_{1e} = \bar{y} \\
\mathbb{E}[Y^2] &= \int_{\theta_1-\theta_2}^{\theta_1+\theta_2} y^2 \frac{1}{2\theta_2} dy = \frac{1}{2\theta_2} \int_{\theta_1-\theta_2}^{\theta_1+\theta_2} y^2 dy = \frac{1}{2\theta_2} \left[\frac{y^3}{3}\right]_{\theta_1-\theta_2}^{\theta_1+\theta_2} \\
&= \frac{1}{2\theta_2} \left(\frac{\theta_1^3 + 3\theta_1^2\theta_2 + 3\theta_1\theta_2^2 + \theta_2^3}{3} - \frac{\theta_1^3 - 3\theta_1^2\theta_2 + 3\theta_1\theta_2^2 - \theta_2^3}{3} \right) \\
&= \frac{1}{2\theta_2} \left(\frac{6\theta_1^2\theta_2 + 2\theta_2^3}{3} \right) = \theta_1^2 + \frac{\theta_2^2}{3} = \mathrm{Var}(Y) + \mathbb{E}[Y]^2 \\
&\implies \mathrm{Var}(Y) = \frac{\theta_{2e}^2}{3} \qquad \implies \theta_{2e} = \sqrt{3s_y^2}
\end{aligned}$$

Where $\bar{y} = \sum_{i=1}^n y_i/n$ and $s_y^2 = \sum_{i=1}^n (y_i - \bar{y})^2/n$. Now we calculate θ_{1e} and θ_{2e} for the given sample:

$$\begin{aligned}
\theta_{1e} &= \frac{8.3 + 4.9 + 2.6 + 6.5}{4} = 5.575 \\
\theta_{2e} &= \sqrt{3 \cdot \frac{(8.3 - 5.575)^2 + (4.9 - 5.575)^2 + (2.6 - 5.575)^2 + (6.5 - 5.575)^2}{4}} \approx 3.63
\end{aligned}$$

Problem 4.

You need to do the MLE in 5.2.13 as well to make a comparison.

4.1 5.2.13 If the random variable Y denotes an individual's income, Pareto's law claims that $\mathbb{P}(Y \geq y) = \left(\frac{k}{y}\right)^{\theta}$, where k is the entire population's minimum income. It follows that $F_Y(y) = 1 - \left(\frac{k}{y}\right)^{\theta}$, and, by differentiation,

$$f_Y(y;\theta) = \theta k^{\theta}\left(\frac{1}{y}\right)^{\theta+1}, \quad y \geq k; \quad \theta \geq 1$$

Assume k is known. Find the maximum likelihood estimator for θ if income information has been collected on a random sample of 25 individuals.

Solution:

$$\ell(\theta) = \sum_{i=1}^{n} \log f_Y(y_i;\theta) = \sum_{i=1}^{n} \log\left(\theta k^{\theta}\left(\frac{1}{y_i}\right)^{\theta+1}\right) = n\log\theta + n\theta\log k - (\theta+1)\sum_{i=1}^{n}\log y_i$$

$$\frac{d}{d\theta}\ell(\theta) = \frac{d}{d\theta}\left(n\log\theta + n\theta\log k - (\theta+1)\sum_{i=1}^{n}\log y_i\right) = \frac{n}{\theta} + n\log k - \sum_{i=1}^{n}\log y_i = 0$$

$$\implies \hat{\theta} = \frac{n}{\sum_{i=1}^{n}\log y_i - n\log k} \implies \hat{\theta} = \frac{25}{\sum_{i=1}^{25}\log y_i - 25\log k}$$

4.2 5.2.22 Find a formula for the method of moments estimate for the parameter θ in the Pareto pdf,

$$f_Y(y;\theta) = \theta k^{\theta}\left(\frac{1}{y}\right)^{\theta+1}, \quad y \geq k; \quad \theta \geq 1$$

Assume that k is known and that the data consist of a random sample of size n. Compare your answer to the maximum likelihood estimator found in Question 5.2.13.

Solution:

$$\mathbb{E}[Y] = \int_k^{\infty} y\theta k^{\theta}\left(\frac{1}{y}\right)^{\theta+1} dy = \theta k^{\theta}\int_k^{\infty}\left(\frac{1}{y}\right)^{\theta} dy = \theta k^{\theta}\left[-\frac{1}{\theta-1}\left(\frac{1}{y}\right)^{\theta-1}\right]_k^{\infty} = \frac{\theta k}{\theta-1} = \bar{y}$$

$$\implies \theta k = \bar{y}(\theta-1) \implies \theta_e = \frac{\bar{y}}{\bar{y}-k}$$

Where $\bar{y} = \sum_{i=1}^{n} y_i/n$. Comparing this with the MLE, the MoM estimate relies on $\bar{y}$ while the MLE relies on the sample logs. These are different and generally won't give the same results.

Problem 5. 5.2.4

Suppose a random sample of size n is drawn from the probability model

$$p_X(k;\theta) = \frac{\theta^{2k}e^{-\theta^2}}{k!}, \quad k = 0, 1, 2, \ldots$$

Find a formula for the maximum likelihood estimator, $\hat{\theta}$.

Solution:

$$\ell(\theta) = \sum_{i=1}^{n} \log p_X(k_i;\theta) = \sum_{i=1}^{n} \log\left(\frac{\theta^{2k_i}e^{-\theta^2}}{k_i!}\right) = \sum_{i=1}^{n}(2k_i \log\theta - \theta^2 - \log k_i!)$$

$$\frac{d}{d\theta}\ell(\theta) = \frac{d}{d\theta}\left(\sum_{i=1}^{n}(2k_i \log\theta - \theta^2 - \log k_i!)\right) = \frac{2}{\theta}\sum_{i=1}^{n} k_i - 2\theta n = 0 \implies \hat{\theta} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} k_i} = \sqrt{\bar{k}}$$

Problem 6. 5.2.6

Use the method of maximum likelihood to estimate θ in the pdf

$$f_Y(y;\theta) = \frac{\theta}{2\sqrt{y}}e^{-\theta\sqrt{y}}, \quad y \geq 0$$

Evaluate θ_e for the following random sample of size 4: $y_1 = 6.2, y_2 = 7.0, y_3 = 2.5,$ and $y_4 = 4.2$.

Solution:

$$\ell(\theta) = \sum_{i=1}^{n} \log\left(\frac{\theta}{2\sqrt{y_i}}e^{-\theta\sqrt{y_i}}\right) = n\log\theta - \log 2 - \frac{1}{2}\sum_{i=1}^{n}\log y_i - \theta\sum_{i=1}^{n}\sqrt{y_i}$$

$$\frac{d}{d\theta}\ell(\theta) = \frac{d}{d\theta}\left(n\log\theta - \log 2 - \frac{1}{2}\sum_{i=1}^{n}\log y_i - \theta\sum_{i=1}^{n}\sqrt{y_i}\right) = \frac{n}{\theta} - \sum_{i=1}^{n}\sqrt{y_i} = 0 \implies \hat{\theta} = \frac{n}{\sum_{i=1}^{n}\sqrt{y_i}}$$

$$\implies \theta_e = \frac{4}{\sqrt{6.2} + \sqrt{7.0} + \sqrt{2.5} + \sqrt{4.2}} \approx 0.456$$

Problem 7.

Suppose $X_1, \ldots, X_n$ is a random sample from the uniform distribution on $[a, b]$.

7.1 Find the method of moments estimators of a and b.

Solution:

$$\mathbb{E}[X] = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx = \frac{1}{b-a} \left[\frac{x^2}{2}\right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2} = \bar{X} \implies \hat{a} = 2\bar{X} - \hat{b}$$

$$\mathbb{E}[X^2] = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx = \frac{1}{b-a} \left[\frac{x^3}{3}\right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3} = \mu_2$$

$$\implies \mu_2 = \frac{(2\bar{X} - \hat{b})^2 + (2\bar{X} - \hat{b})\hat{b} + \hat{b}^2}{3} = \frac{\hat{b}^2 - 2\bar{X}\hat{b} + 4\bar{X}^2}{3} = \frac{(\hat{b} - \bar{X})^2 + 3\bar{X}^2}{3}$$

$$\implies \hat{b} = \bar{X} + \sqrt{3(\mu_2 - \bar{X}^2)}$$

$$\implies \hat{a} = \bar{X} - \sqrt{3(\mu_2 - \bar{X}^2)}$$

Where $\bar{X} = \sum_{i=1}^n X_i / n$.

7.2 Find the maximum likelihood estimators of a and b.

Solution:

$$L(a, b) = \prod_{i=1}^n f(X_i; a, b) = \begin{cases} \frac{1}{(b-a)^n} & \forall X_i \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

To maximize this, you need to minimize $(b-a)$ such that all X_i are in $[a, b]$. Thus, $a = \min_{i=1}^n X_i$ and $b = \max_{i=1}^n X_i$.